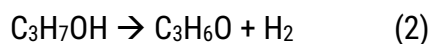
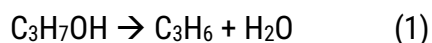


Worksheet: Mass and mole balances on a reactor

Name(s): _____

Fill in all sections – These are today's notes.

In this experiment n-propanol liquid is fed to a steady-state reactor system. The n-propanol vaporizes before entering the reactor where two reactions take place:

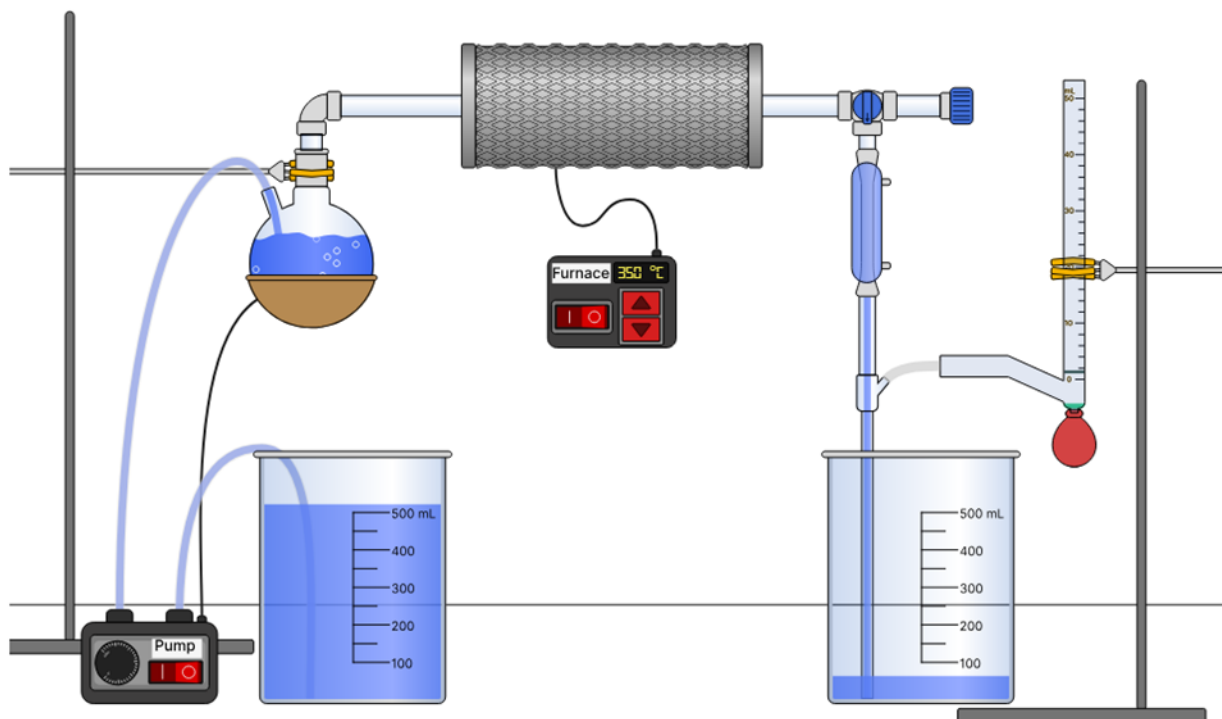


Some n-propanol ($\text{C}_3\text{H}_7\text{OH}$) reacts to propylene (C_3H_6) and water (reaction 1), and some reacts to produce $\dot{n}_{\text{propanol}}$ ($\text{C}_3\text{H}_6\text{O}$) and hydrogen (reaction 2). The remaining n-propanol leaves the reactor without reacting. The water, propanal, and unreacted n-propanol leave the system in the liquid phase. The hydrogen and propylene leave in the gas phase.

Student Learning Objectives

1. Write steady-state material balances around different control volumes in a process.
2. Use reaction stoichiometry to relate reactant consumption to product formation.
3. Convert liquid and gas volumetric flow rates into mass and molar flow rates.

Equipment



Questions to answer before running the experiment

1. We would like to measure the molar percentage yields of the reaction products propanal and propylene. How will the measurements in this experiment enable you to determine those yields? If reaction produces more propylene than propanal, how will this affect the rate at which liquid accumulates in the product beaker?
2. How might the rates of liquid and gas exiting the reactor change as a function of temperature? If one goes up, must the other go down?
3. To analyze the results, we will assume the reaction is operating at steady state. However, there may be an undesired side reaction that continuously produces solid species that accumulate inside the reactor. How would one use the measurements to detect whether this might be a major issue?

Notation

$\dot{n}_{propanol}^0$ = molar flow rate of n-propanol entering reactor (mol/s)

$\dot{m}_{propanol}^0$ = mass flow rate of n-propanol entering reactor (g/s)

$\dot{m}_{out}$ = outlet mass flow rate of the liquid phase (g/s)

$\dot{m}_{propanol}$ = mass flow rate of unreacted n-propanol leaving the reactor (g/s)

$\dot{n}_{propanol}$ = molar flow rate of unreacted n-propanol leaving the reactor (mol/s)

$\dot{m}_{water}$ = mass flow rate of water leaving reactor (g/s)

$\dot{m}_{C_3H_6O}$ = mass flow rate of propanol leaving the reactor (g/s)

$\dot{n}_{gas}$ = gas molar flow rate leaving reactor (mol/s)

$\dot{m}_{H_2}$ = mass flow rate of H₂ leaving reactor (g/s)

$\dot{m}_{propylene}$ = mass flow rate of propylene leaving reactor (g/s)

$MW_{C_3H_6O}$ = molecular weight of propanal (58.1)

$MW_{propanol}$ = molecular weight of n-propanol (60.1)

$MW_{propylene}$ = molecular weight of propylene (42.1)

x = fraction of reacted n-propanol that produced propanal

$1 - x$ = fraction of reacted n-propanol that produced water

Running the experiment

1. Set the 3-way valve at the reactor exit to the exhaust.
2. Turn on the reactor furnace (I is on) and set the reactor temperature to 300°C.
3. Turn on the pump (I is on), which also turns on the evaporator heater, and turn the pump dial to select a feed flow rate of n-propanol to the evaporator.
4. Allow the system to reach steady state.
5. Turn the 3-way valve so the reactor effluent flows to the condenser.
6. Start a timer to measure the change in volume of the feed beaker and record measurements in Table 1.
7. Once liquid starts collecting in the beaker, start a timer and measure the change in volume of liquid collected in the beaker and record measurements in Table 2.
8. Click and hold on the red bulb of the bubble flow meter until a “bubble” move up the meter. Then release holding the bulb. Use a timer to measure the volumetric flow rate of gas leaving the reactor and record measurements in Table 2.
9. You may need to “reset beakers” to complete measurements of the liquid and vapor flow rates out of the reactor.
10. Repeat the measurements of liquid and vapor flow rates to obtain average values.
11. Set the 3-way valve at the reactor exit to the exhaust.
12. Click “reset beakers.”
13. Set the reactor temperature to 330°C.
14. Repeat steps 4 – 12 at 330°C.

Table 1 Liquid volumetric feed rates

Temperature	Feed liquid volume change (mL)	Elapsed time (s)	Feed liquid volumetric flow rate (mL/s)
300°C			
300°C			
300°C			
300°C average			
330°C			
330°C			
330°C			
330°C average			

Table 2 Outlet volumetric flow rates

Temperature	Product liquid volume change (mL)	Elapsed time (s)	Product liquid volumetric flow rate (mL/s)	Vapor volume change (mL)	Elapsed time (s)	Gas volumetric flow rate (mL/s)
300°C						
300°C						
300°C						
300°C average						
330°C						
330°C						
330°C						
330°C average						

15. Use the data in Tables 1 and 2 to compute average mass flow rates of liquid out of the feed beaker. Also compute average mass flow rates into the product beaker. The liquid density is 0.8 g/mL. Enter the values in Table 3. Use the data in Table 2 to compute average gas molar flow rates using the ideal gas law and enter the values in Table 3. The system is at 1.0-bar pressure.

Table 3 Liquid mass flow rates and gas molar flow rates

Temperature	Feed liquid volumetric flow rate (mL/s)	Feed liquid mass flow rate (g/s)	Product liquid volumetric flow rate (mL/s)	Product liquid mass flow rate (g/s)	Gas volumetric flow rate (mL/s)	Gas molar flow rate (mol/s)
300°C average						
330°C average						

Data analysis:

- Record the feed mass flow rates of propanol ($\dot{m}_{propanol}^0$) in Table 4 and solve for the molar flow rates of propanol entering the reactor ($\dot{n}_{propanol}^0$).
- Record the molar flow rates of gas exiting the reactor in Table 4. Each time one mole of propanol reacts, one mole of gas (either hydrogen or propylene) exits the reactor.

$$\dot{n}_{propanol}^0 - \dot{n}_{propanol} = \dot{n}_{gas}$$

Use this equation to calculate the molar flow rates of n-propanol leaving the reactor and record the values in Table 4. Use the molar weight of n-propanol to calculate the mass flow rates of propanol exiting the reactor and record these values in Table 4.

Table 4 Propanol mass and molar flow rates

Temperature	$\dot{m}_{propanol}^0$ (g/s)	$\dot{n}_{propanol}^0$ (mol/s)	$\dot{n}_{gas}$ (mol/s)	$\dot{n}_{propanol}$ (mol/s)	$\dot{m}_{propanol}$ (g/s)
300°C					
330°C					

3. Determine the composition of the liquid stream exiting the condenser at each temperature. The remaining mass accumulating in the product beaker (after subtracting the propanol mass flow rate exiting the reactor) is due to propanol and water. Use stoichiometry to determine the mass flow rates of propanol and water: each time a mole of propanol reacts, it produces a mole of water (which weighs 18 grams) or a mole of propanal (which weighs 58 grams). The outlet mass flow rate of the liquid phase $\dot{m}_{out}$ is:

$$\dot{m}_{out} = \dot{m}_{propanol} + \dot{m}_{C_3H_6O} + \dot{m}_{water} \quad [1]$$

where $\dot{m}_{propanol}$ has already been determined. The mass flow rates of C_3H_6O and water can be determined by relating them to the fraction of propanol that reacted through a given pathway. For example:

$$\dot{m}_{C_3H_6O} = x(\dot{n}_{propanol}^0 - \dot{n}_{propanol})MW_{C_3H_6O} \quad [2]$$

where x is the fraction of reacted propanol that produced propanal. Thus, the quantity $(1 - x)$ is the fraction of reacted propanol that produces water.

A related expression can be written for the mass flow rate of water. Thus, equation [1] can be expressed in terms of a single unknown x , which can be solved for to determine the liquid composition.

$$\dot{m}_{water} = (1 - x)(\dot{n}_{propanol}^0 - \dot{n}_{propanol})MW_{water}$$

4. The quantity $\frac{x(\dot{n}_{propanol}^0 - \dot{n}_{propanol})}{\dot{n}_{propanol}^0}$ is one measure of the fractional molar yield of propanal, i.e., the fraction of the fed reactant that ends up being directed toward the propanal-producing reaction. Determine the fractional molar yields of propanal and propylene as a function of temperature and record the values in Table 5.

Table 5 Fractional molar yields

Temperature	Fractional molar yield	
	Propanal	Propylene
300°C		
330°C		

5. Compute a total mass balance on the entire system. In other words, determine the mass flow rate into the reactor, and then sum up the mass flow rates of all compounds exiting the reactor. (This requires determining the mass flow rate of the vapor stream!) Are the inlet and outlet mass flow rates equal? If they are not (or if they were not), what would this tell you physically?

Table 6 Mass flow rates (g/s) into and out of reactor

	In	Out						In-Out
Temperature	$\dot{m}_{propanol}^0$	$\dot{m}_{propanol}$	$\dot{m}_{C_3H_6O}$	$\dot{m}_{propylene}$	$\dot{m}_{water}$	$\dot{m}_{H_2}$	Total	
300°C								
330°C								

Questions to answer

1. Which single measurement has the greatest effect on the calculated product yields? Explain why.
2. How does reactor temperature affect the outlet gas flow rate? Why do you think it has the effect it does?
3. The reactor furnace must supply heat to vaporize the n-propanol feed and maintain the reactor temperature. What energy terms would appear in an energy balance around the reactor?
4. What are sources of error in the measurements? What could cause mass balances to be in error?
5. What safety measures would you employ if making this measurement in the laboratory?